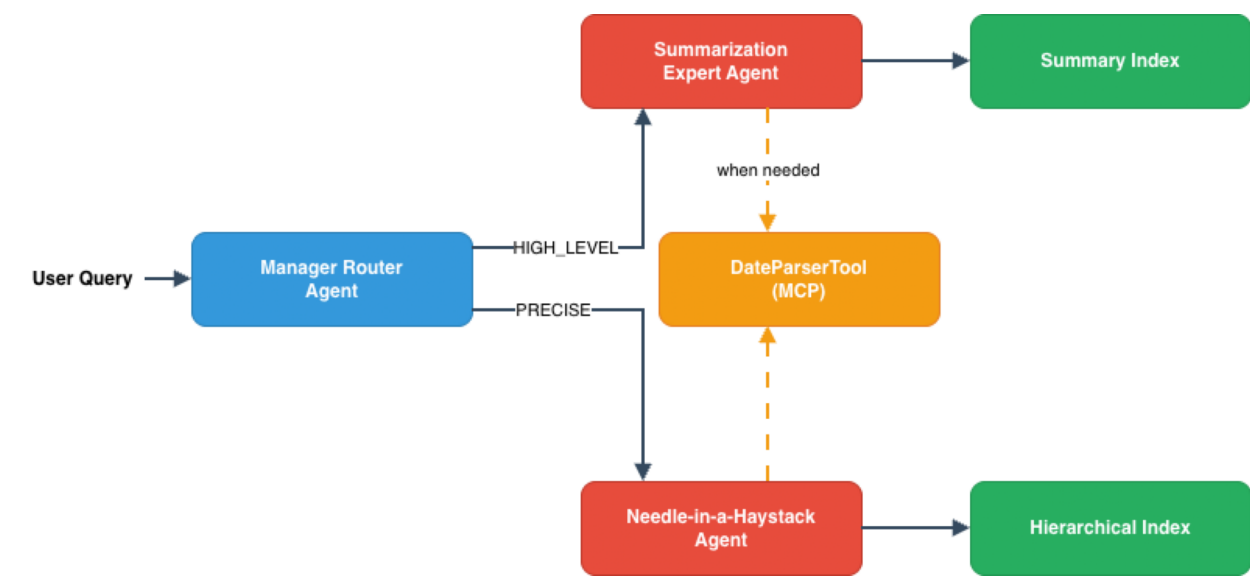


Multi-Agent System with MCP Integration

System Overview

A multi-agent system using LlamaIndex for insurance claim analysis with intelligent routing, specialized indexes, and MCP (Model Context Protocol) tool integration. The system implements a three-agent architecture: Manager Router Agent classifies queries as HIGH_LEVEL or PRECISE, routing to specialized agents (Summarization Expert or Needle-in-a-Haystack) that use appropriate indexes (Summary Index with large chunks or Hierarchical Index with multi-size chunks).

Architecture



Manager Router Agent uses LLM-based classification with keyword fallback. Summarization Expert handles high-level queries using `tree_summarize` (`top_k=15`) with large chunks (800-1200 tokens). Needle-in-a-Haystack Agent handles precise queries using compact mode (`top_k=30`) with multi-size chunks. Both agents can use `DateParserTool` (MCP) for date parsing when needed. Storage: ChromaDB (PersistentClient) stores all vector embeddings in `chroma_db/` directory, enabling persistence between sessions. Indexes are automatically loaded from ChromaDB on startup if they exist, otherwise created from documents. Local storage (`storage/`) contains `hierarchy_info.json` with metadata about claims, documents, and sections structure.

Main Results

Evaluation Metrics (11 test cases):

- Overall Score: 0.85 (Correctness: 0.77, Relevancy: 0.99, Recall: 0.83)
- Routing Accuracy: 100.0% (correct index selection)
- Index Selection Accuracy: 100.0%

Performance by Category:

- High-Level Summary: Excellent performance (0.94-1.00) with comprehensive summaries
- Precise Factual: Good performance (0.44-1.00), challenges with rare information
- High-Level Categorization: Perfect performance (1.00)

MCP Usage

`DateParserTool` (MCP) extends LLM capabilities beyond text retrieval. It automatically detects date-related keywords (date, time, when, timestamp) and provides: date parsing from various formats, normalization to standard format, extraction from text, and date difference calculations. The tool is available to both agents when date parsing is needed, enabling queries like 'What date was the claim filed?' or 'When was the inspection completed?' without requiring the LLM to parse dates from retrieved text chunks.